

DAY14 — nmap-Basics

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THEORY

1. What does Nmap stand for?

It stands for network mapper.

2. What is Nmap used for?

It is used to do a network discovery so it is used to discover and investigate devices and services on a network

3. What is network discovery?

Discovering and investigating devices and services on a network

4. What is a port scan?

Investigating if a port is open or closed or filtered.

5. What does an open port generally indicate?

It indicates that there is a service listening on this port.

6. Does an open port automatically mean vulnerability? Explain.

No not necessarily it just means that a service is listening on this port so an open port is not an evidence of a vulnerability its just a clue where we can look at open ports and investigate services and see if there a security weakness

7. What does closed mean conceptually?

It means that the service is not listening on a port

8. What does filtered mean conceptually?

It means that nmap can't indicate if a port is open or closed because a firewall or another network mechanism is blocking the scanning

9. What does nmap 127.0.0.1 do?

It scans the localhost which is your own device to identify the reachable tcp ports

10. What does -p do?

It specifies which port or range of ports to scan

11. What does -sV do?

It is used to scan for service and version on ports.

12. What does -oN do?

It saves the nmap results as a human readable file

13. Why should scan results be documented?

They should be documented for investigations and to compare results over time to see if there is something abnormal

14. What is cybersecurity scope?

It is like how far you can go ; what devices , systems, networks , ip addresses you can access for testing.

15. Why must authorization exist before scanning?

Surely because unauthorized access can lead to lots of problems

16. What is the difference between scanning and exploitation?

Scanning is for investigations and making sure the system is secure while exploitation is used to compromise a system or a device or a service

17. Why is localhost a good beginner target?

Because you are testing your own device so it will never be an unauthorized target so you are not getting on trouble

18. How do ports relate to services?

Services use ports where ports act like endpoints for services , a service listens on a port so other systems can communicate with it.

19. Why are unexpected listening services worth reviewing?

Because it means that there is an unwanted activity going either a misconfiguration or an attack so it must be investigated

20. Explain a responsible Nmap assessment workflow.

Define scope and obtain authorization → Identify the target → Perform host discovery → Scan relevant ports → Identify services and versions → Review the results → Document findings → Investigate potential security issues → Report the results.

The most important part is not to try nmap on unauthorized environment

PRACTICAL SCREENSHOTS

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nmap --version · nmap 127.0.0.1 · nmap -p 22,80,443 127.0.0.1 · nmap -sV 127.0.0.1 · nmap -oN day14-scan.txt 127.0.0.1 · cat day14-scan.txt
```

TASK 10 — Mini report:

TARGET: 127.0.0.1 · AUTHORIZED: YES

OPEN PORTS: there is no open ports SERVICES: http,https,ssh (they are not listening)

OBSERVATIONS: the host is up but no services are listening

RECOMMENDATION:there is nothing to investigate further but better to disable unnecessary services to avoid any security weakness

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-1.18  
.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
  
(kali@kali)-[~]  
$ nmap 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:49 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000020s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds  
  
(kali@kali)-[~]  
$ nmap -p 22,80,443 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:50 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000056s latency).  
  
PORT      STATE SERVICE  
22/tcp    closed ssh  
80/tcp    closed http  
443/tcp   closed https  
  
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:50 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000010s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds  
  
(kali@kali)-[~]  
$ nmap -oN day14-scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:51 -0400  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 127.0.0.1 -oN day14-scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:52 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000010s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)
```

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ nmap -p 22,80,443 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:50 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000056s latency).  
  
PORT      STATE SERVICE  
22/tcp    closed ssh  
80/tcp    closed http  
443/tcp   closed https  
  
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds  
  
(kali@kali)-[~]  
└─$ nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:50 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000010s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds  
  
(kali@kali)-[~]  
└─$ nmap -oN day14-scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:51 -0400  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds  
  
(kali@kali)-[~]  
└─$ nmap -sV 127.0.0.1 -oN day14-scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 14:52 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000010s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds  
  
(kali@kali)-[~]  
└─$ cat day14-scan.txt  
# Nmap 7.99 scan initiated Fri Sep 25 14:52:43 2026 as: /usr/lib/nmap/nmap --privileged -sV -oN day14-scan.txt 127.0.0.1  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000010s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
# Nmap done at Fri Sep 25 14:52:43 2026 -- 1 IP address (1 host up) scanned in 0.13 seconds  
  
(kali@kali)-[~]  
└─$  
  
(kali@kali)-[~]  
└─$
```